Analyzing Impact of Workload on Injury Risk and Performance in the NFL

2025-26 NFLPA Competition
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1. BACKGROUND

The physicality of the NFL comes along with not only entertainment, but also significant injury risk on the players' bodies. In particular, running backs and wide receivers face high injury rates as a study of lower extremity injuries from 2015-2018 found that wide receivers (WRs) make up 14.4% of such injuries and running backs (RBs) make up 9.3% of such injuries [1].

Since RBs and WRs have well-documented injury risks, we investigated how workload impacts injury risks and performance for these two position groups throughout the NFL season. We focus on the influence of factors such as position, usage, and touch volume on week-to-week efficiency metrics, including yards per touch and expected points added (EPA) per touch, as well as the correlation between workload and the likelihood of missing future games.

2. METHODOLOGY

2.1 Research Question

Our guiding research question was: "How does workload affect injury risk and performance for RBs and WRs over the course of a season?"

2.2 Defining Key Measures

In the context of our research question, we define workload differently for RBs and WRs. For RBs, workload is considered as the number of touches (rushes + receptions) the player has per game. For WRs, workload is considered the number of targets the player has per game. We decided to use these metrics to represent workload because touches and targets represent opportunities for physical contact opportunities which are important in capturing injury risk.

We measured injury risk using games missed, since no reliable injury dataset could be cleanly linked to our week-by-week data, making game participation the most consistent and straightforward indicator. While our method was reliable, a key limitation is that it does not distinguish absences due to coaching decisions and cannot capture in-game injuries or injuries that allow players to continue without missing a full week.

We defined performance using yards per touch and EPA per play for RBs, and yards per reception and EPA per play for WRs. We chose a mix of traditional and advanced statistics to better reflect both player efficiency and on-field impact.

2.3 Hypothesis

Based on our research question and defined key measures, our hypothesis was that players who took on heavier workloads would have a decline in performance and a greater risk for injuries as the season progressed.

2.4 Data Preparation & Feature Engineering

The first step involved obtaining our data, which we did using the Python package nflreadpy [2]. Nflreadpy enables downloading NFL data across multiple domains, including weekly statistics, contract information, and team schedules. Specifically, we used nflreadpy to collect week-by-week data for all RBs and WRs from the 2021–2024 seasons, along with corresponding team schedule information. The 2021–2024 window was selected because it represents the complete span of seasons played under the NFL's 18-week format.

Using the data obtained through nflreadpy, we created a set of features relevant to answering our research question. The following list goes more into greater detail about the engineered features derived from nflreadpy data. The script used to generate the features can be found in Appendix A.

1. Quartile: based on Weeks 1-4 avg touches/targets (Q4 = high workload; Q1 = low workload)
2. Touches: Tracks average efficiency metrics by week for high-workload players (Q3 & Q4). For RBs, we analyze yards per touch & EPA; for WRs, we analyze yards per reception and EPA. Tests whether performance declines over the season due to cumulative fatigue.
3. Touch_bin / Target_bin: categorical bins for touches/targets
4. Total_epa: receiving_epa + rushing_epa
5. Yards_per_touch / Yards_per_reception: rushing_yards + receiving_yards divided by touches / receiving_yards / receptions
6. Is_bye: true if bye_week matches current week
7. Played: true if player had at least 1 touch or target
8. Is_injury: true if player did not play and it is not a bye week

2.5 Graph Creation

To investigate the relationship between workload for injury risk and performance, we developed five distinct visualizations for both RBs and WRs. The visualizations were created using the Matplotlib and Seaborn libraries in Python [3, 4]. Below is a brief description of each visualization. Code for visualizations is provided in Appendices B–F.

3. RESULTS

3.1 Graphs

3.1.1 Survival Curve

The survival curve measures the percentage of players in each workload quartile still playing in each week of the season and assesses whether high-workload players are more likely to miss games as the season progresses.

Figure 1. Percentage of RBs still active each week by initial workload quartile.

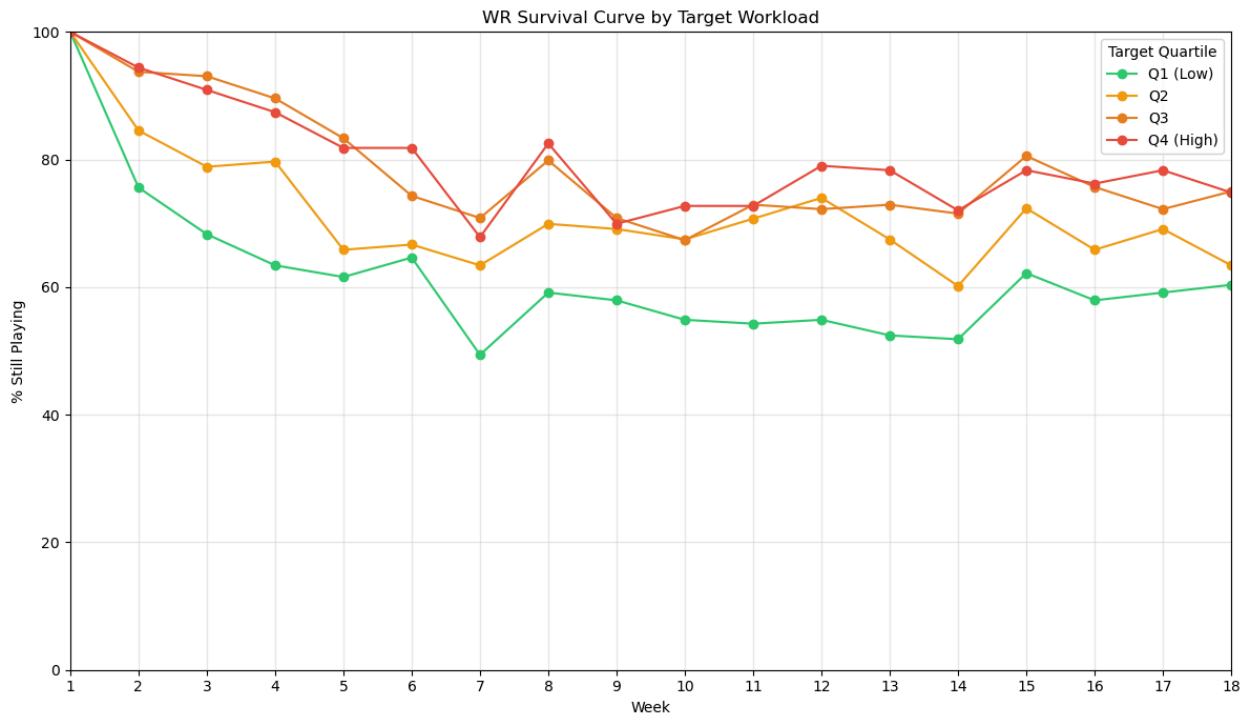

Figure 2. Percentage of WRs still active each week by initial workload quartile.

3.1.2 Efficiency Decline

This graph shows the weekly average efficiency metrics for high-workload players (Q3 & Q4). For RBs, it shows weekly yards per touch and EPA while for WRs it shows weekly yards per reception and EPA. It addresses whether performance declines over the season due to cumulative fatigue.

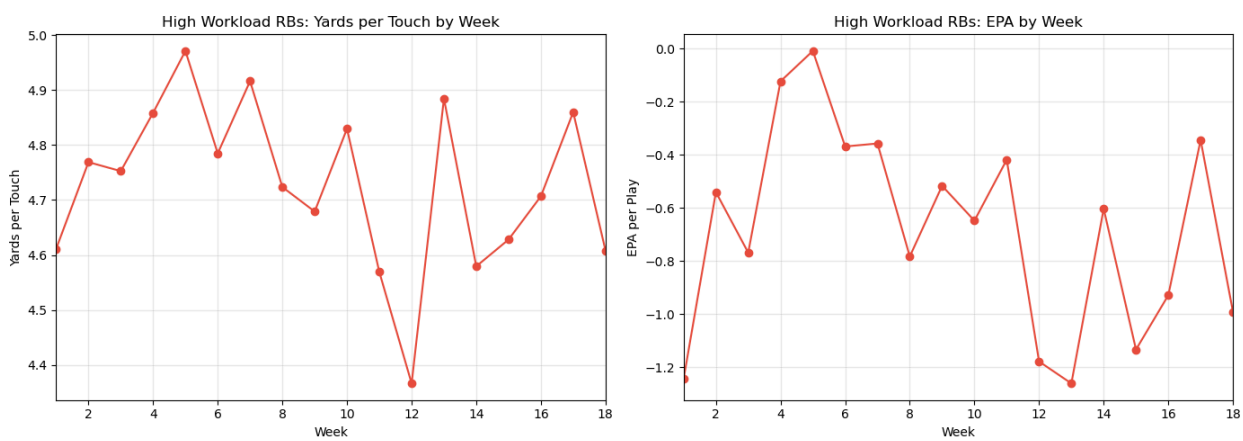

Figure 3. Weekly yards per touch and EPA trends for high-workload RBs (Q3 & Q4) throughout the season.

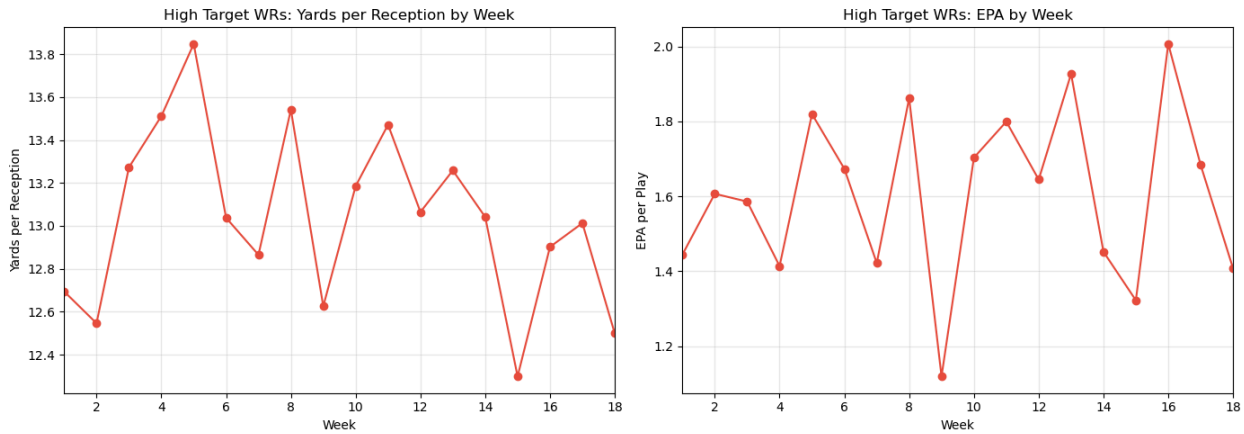

Figure 4. Weekly yards per touch and EPA trends for high-workload WRs (Q3 & Q4) throughout the season.

3.1.3 Injury Risk Heatmap

This graph shows weekly injury rates by workload bin (e.g., 10–15 touches, 15–20 touches, 20+ touches). Each cell represents the percentage of players in that bin who were injured the following week. It assesses whether specific workloads are associated with increased injury risk.

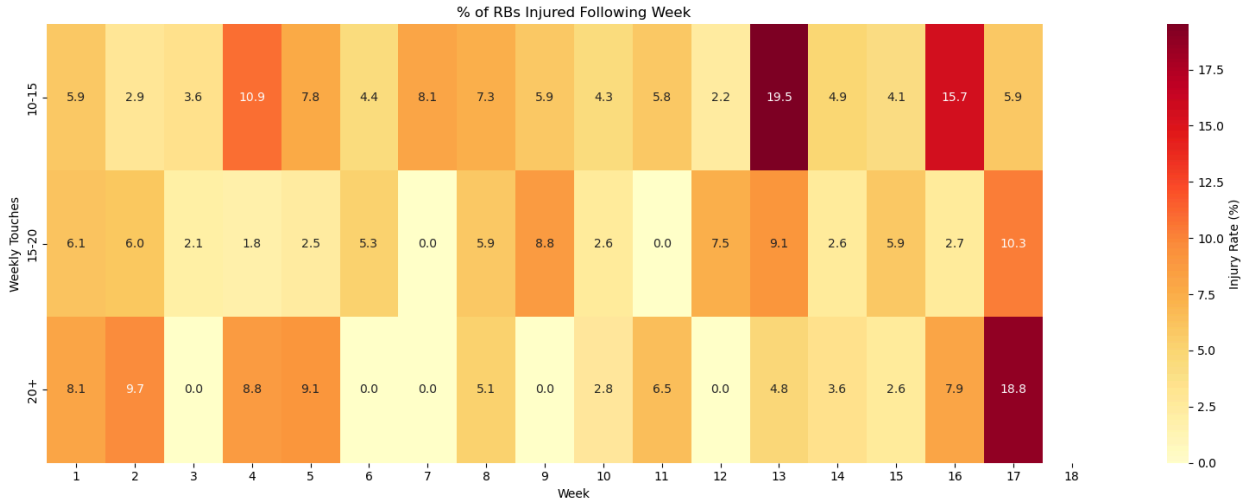

Figure 5. Injury rates for RBs the following week by current week and touch volume.

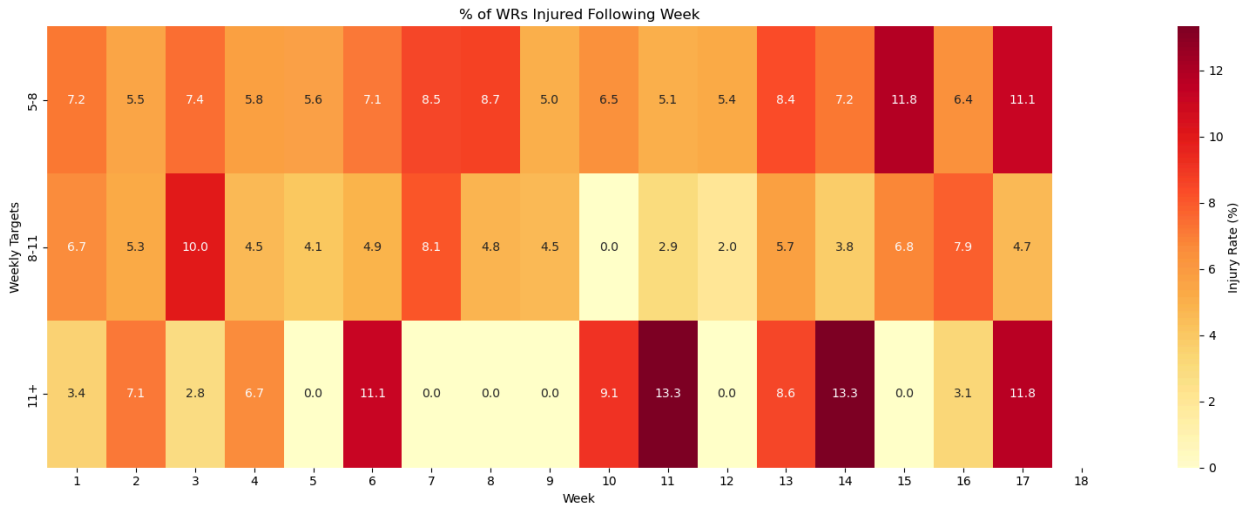

Figure 6. Injury rates for WRs the following week by current week and target volume.

3.1.4 Games Missed Distribution

This graph compares the distribution of consecutive games missed for high-workload players (Q3 & Q4). It assesses whether high-workload players experience longer injury absences when injured.

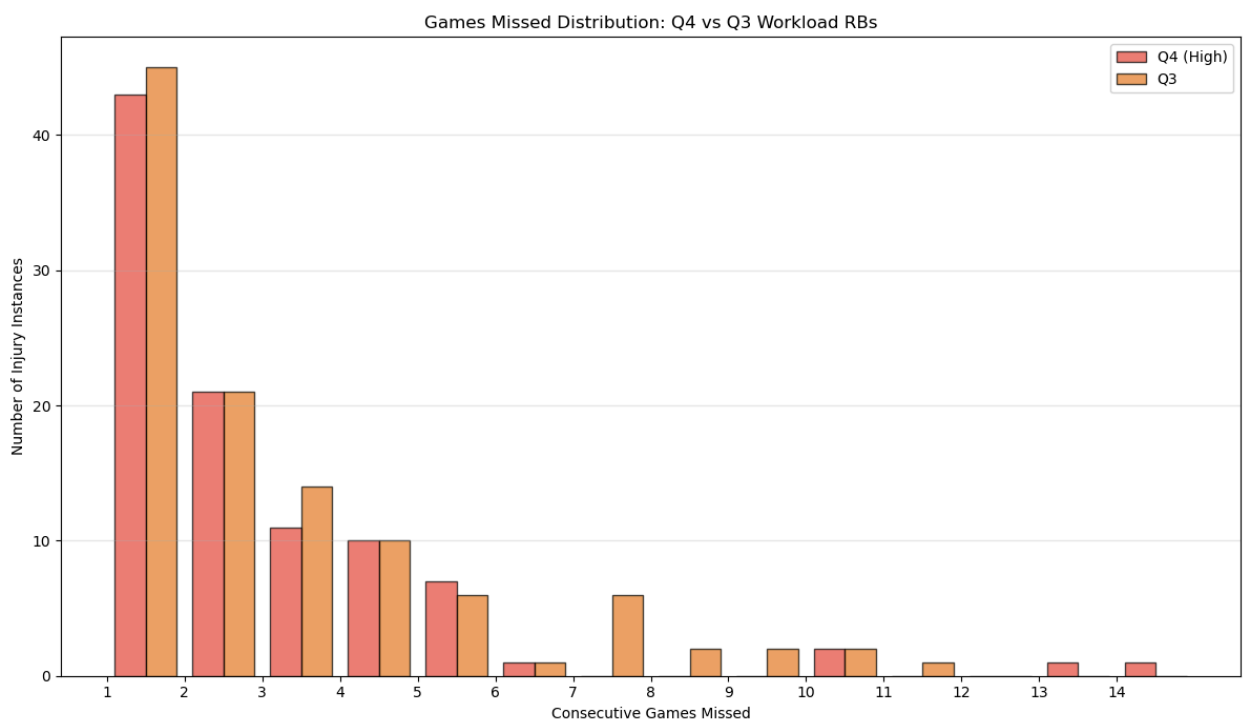

Figure 7. Distribution of consecutive games missed for high-workload (Q3 & Q4) RBs.

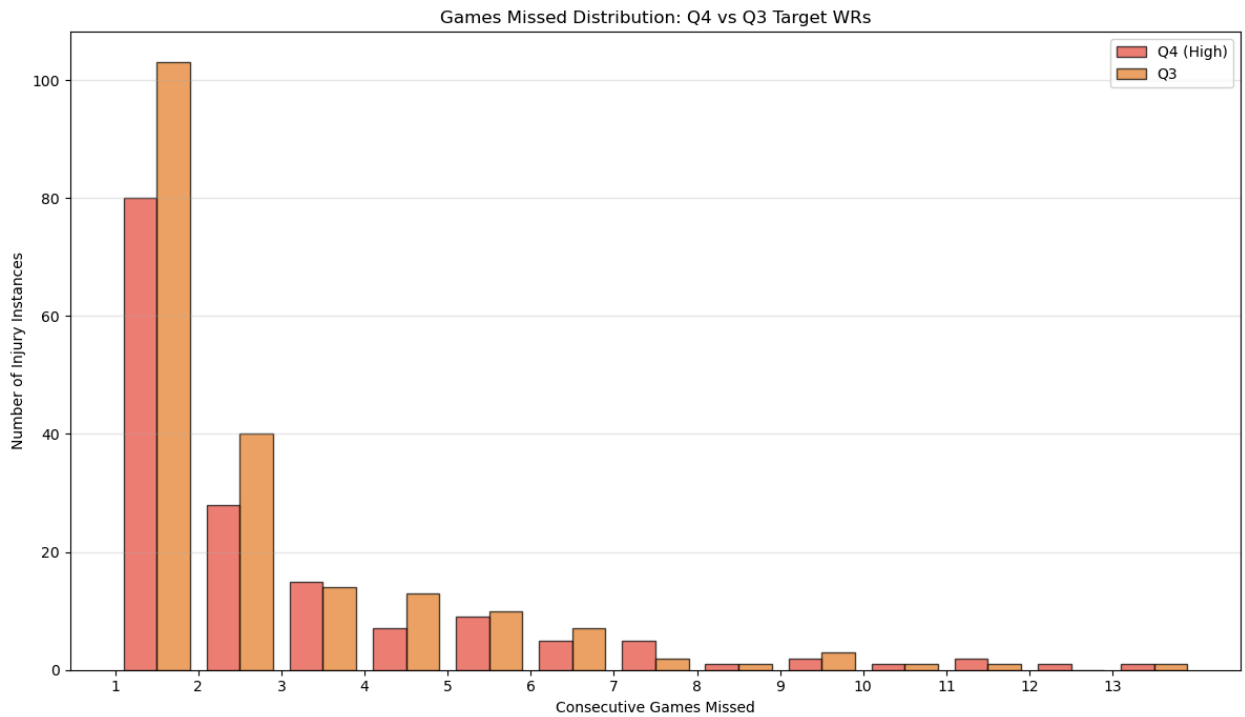

Figure 8. Distribution of consecutive games missed for high-workload (Q3 & Q4) WRs.

3.1.5 Weekly Workload Trends

This graph shows weekly touches or targets for each workload quartile throughout the season. It evaluates whether teams adjust workloads for different player groups as the season progresses.

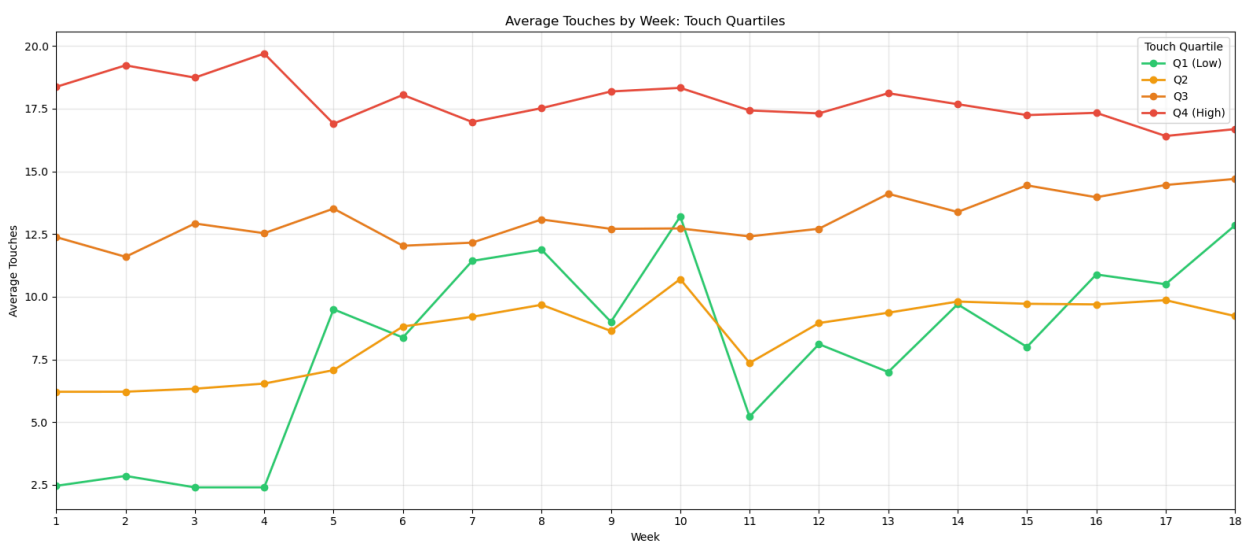

Figure 9. Average touches per week for RBs across the four workload quartiles through the season.

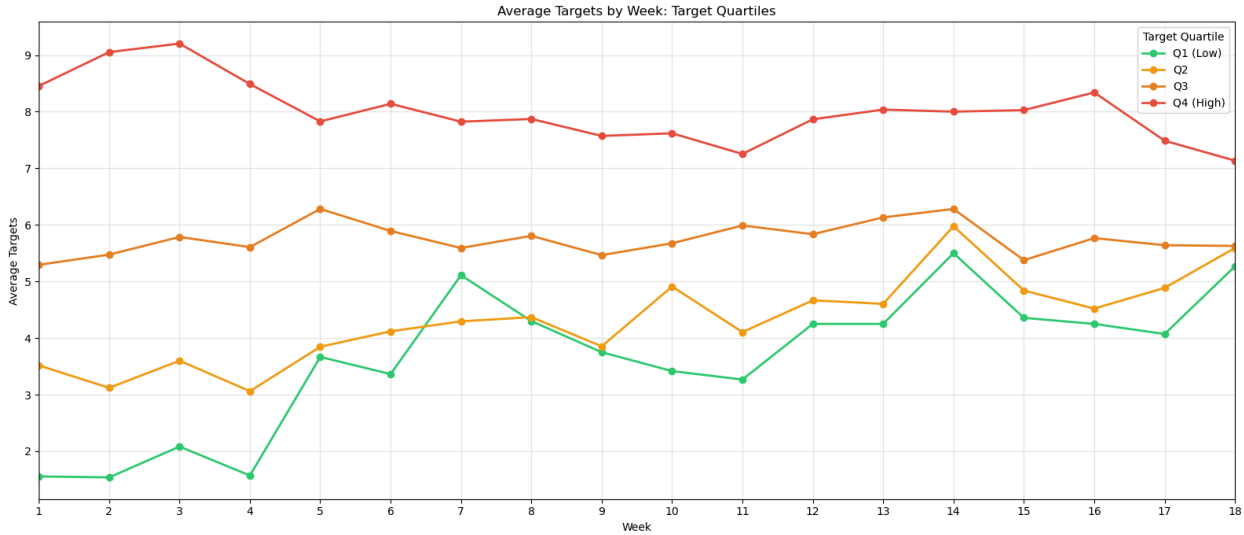

Figure 10. Average touches per week for RBs across the four workload quartiles through the season.

3.2 Statistical Analysis

To assess the patterns observed in our visualizations, we performed statistical tests in R that fit the structure of each graph and the specific outcome it was intended to show. Our statistical tests included linear mixed-effect models, ANOVA, and chi-square tests with a significance threshold of 0.05. The following list goes more into greater detail about the statistical tests used for each graph. Code for the statistical tests is provided in Appendices G–K.

1. Survival Curve: Tested using linear mixed-effect models assessing changes in availability by week, workload quartile, and their interaction, with quartile-level random intercepts.
2. Efficiency Decline: Tested using linear regression to estimate weekly trends in efficiency metrics and assess statistical significance.
3. Injury Risk Heatmap: Tested using one-way ANOVA to evaluate differences in next-week injury probability across workload bins.
4. Games Missed Distribution: Tested using chi-squared analysis to compare distributions of consecutive games missed across workload groups.
5. Weekly Workload Trends: Tested using two-way ANOVA to assess the effects of workload quartile, week, and their interaction on games missed over the season.

3.2.1 Linear Mixed-Effect Models for Survival Curve

RB snap share declines slightly (-0.69% per week), but that is statistically insignificant. Snap participation does not decline meaningfully across workload quartiles; high-workload RBs don't experience steeper decline than low-workload RBs.

WR snap share declines significantly (-1.27% per week) and is statistically significant. However, the rate of decline does not differ by workload quartile.

3.2.2 Linear Regression Test for Efficiency Decline

RB and WR efficiency remains stable throughout the season. The metrics for both position groups (yards/touch, EPA/touch, yards/reception, EPA/play) do not change significantly.

3.2.3 One-way ANOVA for Injury Risk Heatmap

Next-week injury probability does not differ significantly across touch-volume bins for both RBs and WRs.

3.2.4 Chi-square Test for Games Missed Distribution

The distribution of consecutive games missed does not differ significantly between high-workload both RB and WR quartiles.

3.2.5 Two-way ANOVA for Weekly Workload Trends

Trends in games missed do differ between workload quartiles. Low- and moderate-workload RBs (Q1-Q3) show significant increases in games missed, while high-workload RBs show a small but significant decrease. This could be due to survivor bias or durability effects.

For WRs, weekly trends in games missed vary significantly across workload quartiles. Low- and moderate-workload WRs (Q1-Q3) gradually miss more games over the season, whereas high-workload WRs (Q4) show a small but significant decline in games missed, indicating that heavy usage does not translate to increasing absences.

4. CONCLUSION

4.1 Summary of Findings

Based on our visualizations and statistical tests, we rejected our initial hypothesis that players with heavier workloads experienced performance declines or an increased risk of injury as the season progressed. We found that players who took on heavier workloads did not have a significant decline in performance or a significantly greater risk for injuries as the season progressed.

4.2 Implications & Recommendations

Based on these findings, one should avoid assuming a straightforward relationship between increased workload and increased injury risk. In developing strategies to protect players from cumulative workload impacts, it is important to recognize that injury risk is influenced by multiple factors beyond simple measures of performance. Therefore, adopting a more nuanced, individualized approach can provide a more accurate assessment of risk, inform safer practice and game participation decisions, and support long-term player health. To achieve this, the

NFLPA should focus on implementing two key priorities: enhanced workload tracking/individualized monitoring and improved medical communication for players.

For the first priority, the union could advocate for systems that track players' daily and weekly workloads in more detail, including the number of plays, practice intensity, types of drills, and rest periods, with adjustments based on position, prior injuries, and recovery needs. The CBA [5] already sets limits on practice time, padded practices, and mandatory days off (Articles 23 & 35), which provide a basic framework. Building on these rules, the NFLPA could use real workload data to help teams manage players more safely and individually.

For the second priority, the NFLPA should focus on enhancing how players receive, interpret, and act on their medical information. While Article 40 guarantees players access to personnel and medical records and summary reports, the union could advocate for improvements such as faster delivery of records and integration with workload data to inform recovery decisions. These enhancements would allow players to make more informed choices regarding practice participation, recovery, and long-term health planning.

Combined, these two priorities would enable the NFLPA to implement a more data-driven approach to protecting player health. Detailed workload tracking paired with improved medical communication allows for individualized management of training, practice, and game participation, reducing cumulative injury risk and improving overall player well-being. These measures balance player safety with competitive performance, supporting the long-term health of players across the league.

4.3 Limitations

This project is limited by the data available and the structure of the NFL season. Since players are often not played for reasons unrelated to injury, missed games may not perfectly reflect injury severity. Additionally, workload quartiles are created from Weeks 1-4, which can introduce survivorship bias, since the players who remain healthy naturally accumulate higher workloads. The use of descriptive trends and linear models may not capture the non-linear and context-dependent nature of NFL injuries, limiting the precision of our research.

4.4 Future Work

Future research could apply machine learning methods to capture non-linear effects and extend the analysis to additional positions. Examining longer historical windows would also help assess whether these patterns relating to workload have persisted over time.

5. REFERENCES

[1] Mack CD, Kent RW, Coughlin MJ, et al. Incidence of Lower Extremity Injury in the National Football League: 2015 to 2018. The American Journal of Sports Medicine. 2020;48(9):2287-2294. doi:10.1177/0363546520922547

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